

Question 1			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Conversion of eV to J i.e. any 1 of: $0.820 \text{ eV} = 1.312 \times 10^{-19} \text{ J}$ $0.051 \text{ eV} = 0.082 \times 10^{-19} \text{ J}$ $0.769 \text{ eV} = 1.23 \times 10^{-19} \text{ J}$ or by implication (1) Photon energy = $0.82 \text{ eV} - 0.05 \text{ eV}$ or $1.312 \times 10^{-19} \text{ J} - 0.082 \times 10^{-19} \text{ J} = [1.230 \times 10^{-19} \text{ J}]$ or by implication (1) $\lambda = 1617 \text{ [nm]}$ (1) [1516 nm arises from wrong photon energy – 1 mark] Arithmetic slip lose 1 mark	1	1		3	3	
		(ii)	A photon of this wavelength or energy or $1.23 \times 10^{-19} \text{ J}$ or 0.769 eV or 1617 nm (ecf) (accept "correct energy, specific energy") passing an electron in <u>the U state</u> (1) accept "colliding with" ... triggers release of another photon (1) don't accept "photon absorbed and emits two" The photons are in phase or of the same wavelength or energy or frequency or identical or travelling in the same direction (1) don't accept coherent	3			3		
	(b)		Level L self-emptying or short-lived or short time or fall immediately or equivalent (1) Population inversion easier to achieve (accept better or equivalent) or lower pumping [rate] needed (1) don't accept more efficient only	2			2		
			Question 1 total	6	2	0	8	3	0